



Evolving Thematic Map Design in Chinese- and English-Language Academic Cartography: A Three-Decade Data-Driven Study

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Received: 16 June 2026 / Accepted: 1 September 2026
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Abstract

Thematic maps are essential tools for communicating spatial information in academic publications, yet their long-term design evolution across different linguistic publishing contexts remains insufficiently understood. This study presents a large-scale, data-driven analysis of thematic map design in Chinese- and English-language academic cartography from 1990 to 2020. We develop an AI-assisted cartographic data mining framework that integrates vision-language models for understanding heterogeneous academic documents and identifying thematic maps with computer vision techniques for map element detection and image segmentation. Based on this framework, we compile a corpus of 45,732 research articles from 16 representative cartography and geographic information science journals and extract 23,928 high-quality thematic maps for systematic analysis. Map design is characterized along three dimensions: map elements, color design, and layout structure. The results reveal broadly shared cartographic conventions between Chinese- and English-language academic maps, while the degree of convergence varies across design dimensions. English-language maps generally contain slightly richer cartographic elements and greater chromatic diversity, whereas Chinese-language maps show a stronger preference for neutral color schemes. Both groups, however, exhibit highly similar centered and balanced layout structures. Temporal analysis further shows strong convergence in map-element configuration, parallel enrichment of color design with persistent cross-language differences, and comparatively stable layout structures characterized mainly by incremental changes in space allocation and alignment. Overall, academic map design appears to evolve toward shared cartographic conventions while retaining contextual differences associated with publishing traditions.

Keywords Academic cartography · Data-driven analysis · Intelligent cartographic systems · Multimodal large language models · Visual design

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Introduction

Maps have long served as a key medium for communicating geographic knowledge in academic research (White et al. 2017; Coetzee et al. 2021). As scientific publishing continues to expand at an unprecedented pace, cartographic visualizations in academic journals have become increasingly critical for presenting complex spatial analyses and engaging readers (Wu et al. 2024). *Recent geovisualization research has further demonstrated the importance of effective visual representation for supporting the exploration and interpretation of complex multidimensional spatial data* (Gamlen et al. 2025). Reflecting this shift, many journals now place greater emphasis on the visual quality and clarity of figures, and some have introduced more explicit guidelines or professional support for map preparation (Wang et al. 2024a). Consequently, design choices related

to map elements, color usage, and layout structure have a direct impact on how scientific arguments are perceived and understood, and a substantial body of research in cartography has investigated these map design principles.

In terms of map elements, prior studies have examined the selection and visual hierarchy of core components such as the main map and legend, emphasizing their role in conveying structure and importance (Kessler and Slocum 2011). More recently, Jia et al. (2024) and Yang et al. (2026) have summarized the structural elements of the ubiquitous map.

As for color design, a comprehensive set of guidelines grounded in the theory of visual variables was first proposed by Bertin (1974). Then the subsequent works have explored how hue, brightness, and contrast influence perception, readability, and semantic interpretation, leading to widely adopted guidelines for thematic mapping and visual differentiation (e.g., Brewer 1994; MacEachren 2004; Silva et al. 2011; Yang et al. 2025a).

With respect to layout structure, studies have addressed the spatial organization of the main map and auxiliary elements, such as legends and titles, highlighting principles of balance, alignment, and visual flow in page-based map composition (e.g., Dent et al. 2008; Slocum et al. 2009). Together, these works form a well-established foundation for understanding cartographic design. However, much of this knowledge has been derived from expert reasoning, instructional examples, or analyses of relatively small collections of maps.

Recently, the widespread adoption of open-access publishing, together with advances in document analysis and machine learning, has made it increasingly feasible to systematically collect academic maps and automatically extract and analyze their visual components at scale (Zhang et al. 2025). For example, some researchers have focused on scanned map images and thematic content recognition, such as map-type classification and text extraction (e.g., Mandal et al. 2014; Zhou et al. 2018). Subsequent studies expanded toward generative and segmentation-oriented tasks, including image-to-image translation between map styles (Isola et al. 2017), feature extraction from topographic maps (Saeedimoghaddam and Stepinski 2020; Kramm et al. 2025), and layout recognition (Zhou et al. 2024; Yang et al. 2025b). Despite these advances, most existing datasets remain task-driven and are designed to support visual recognition, segmentation, or text detection, with limited attention to cartographic design principles and their empirical evaluation. As a result, large-scale, longitudinal analyses that explicitly target map design characteristics remain largely unexplored.

Motivated by these considerations, this study investigates academic map design through large-scale empirical evidence. We construct an academic map dataset by automatically extracting and analyzing thematic maps from

full-text journal articles using large-model-based document understanding and computer vision techniques (Sect. "Dataset construction"). Based on the extracted maps and design features, we examine cross-language differences and temporal evolution across three dimensions: map elements, color design, and layout structure (Sect. "Design dimensions for analysis"). Specifically, this study addresses two research questions:

- RQ1: Do Chinese- and English-language academic maps exhibit different design patterns in terms of map elements, color usage, and layout structure?
- RQ2: How have academic map design practices evolved over the past three decades, and do Chinese- and English-language maps follow similar or divergent evolutionary trajectories?

To answer these questions, we conduct a systematic analysis of cross-language differences and temporal trends in thematic map design (Sect. "Statistical analysis and results"). The overall analytical framework of the study is illustrated in Fig. 1.

Related Works

Map Design Principles

Map design has been extensively studied in cartography, resulting in a broad body of research on visual, structural, and semantic design principles. Given the scope of this work, we review representative studies from three perspectives that directly correspond to our analysis: (1) map elements, (2) color design, and (3) layout structure.

1. Map elements.

Map elements constitute the fundamental components through which geographic information is organized and communicated, including the main map, legend, scale bar, north arrow, title, annotations, and other auxiliary elements. Classical cartographic research has emphasized that these elements are not merely decorative components but serve distinct cognitive and communicative functions by establishing visual hierarchy, supporting interpretation, and improving map usability (Dent et al. 2008; Slocum et al. 2009). Among these elements, the relationship between the main map and supporting components, particularly legends and annotations, has received considerable attention because they determine how effectively users connect visual representations with geographic meanings (MacEachren 2004). Recent studies have further attempted to characterize

The overall analytical framework

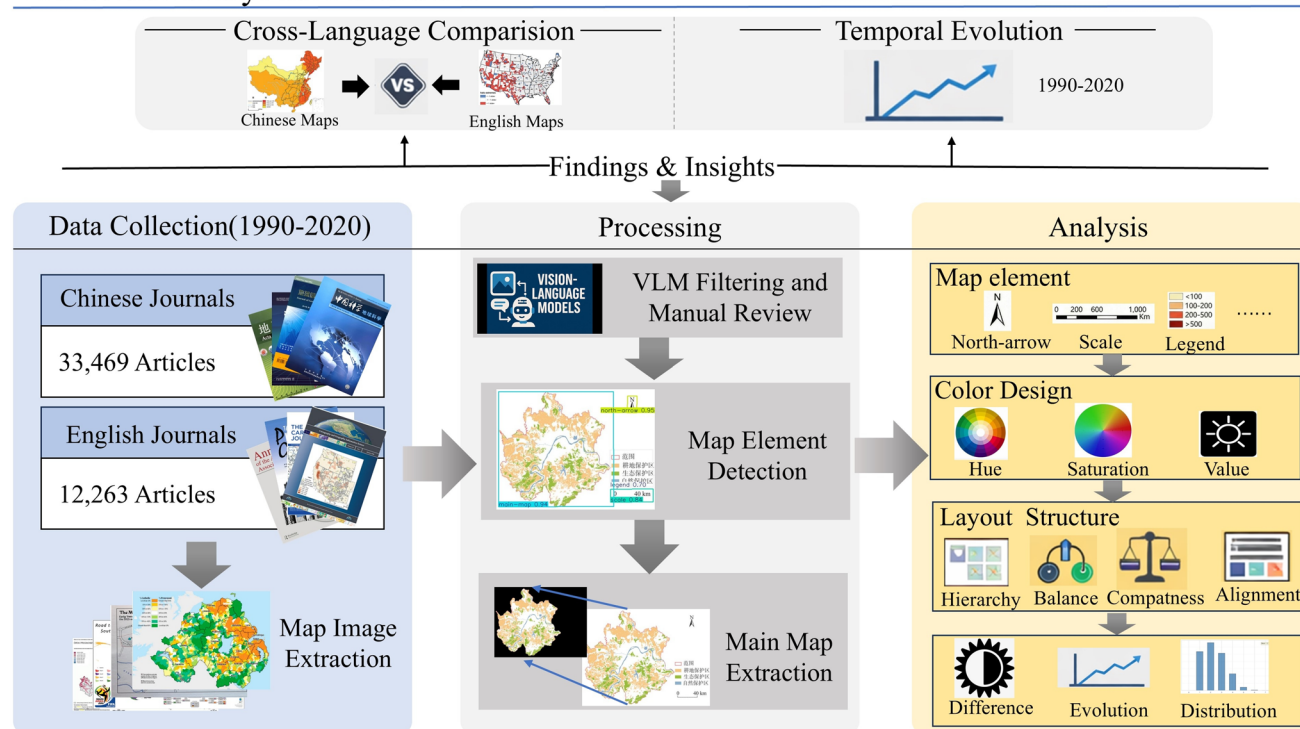


Fig. 1 The overall analytical framework of this study

and quantify map elements from large collections of maps. For example, Jia et al. (2024) and Yang et al. (2026) analyzed common structural components of ubiquitous maps and explored their distribution patterns across different map types. Meanwhile, data-driven approaches have enabled automatic detection and representation of map elements, supporting large-scale map understanding and analysis (Yang et al. 2025b). These advances provide a foundation for analyzing map elements.

2. Color design.

Color design is one of the most thoroughly studied aspects of map design, as color plays a critical role in encoding thematic information and guiding visual perception. Early foundational work by Bertin (1974) introduced the theory of visual variables, identifying hue, brightness, and saturation as fundamental dimensions for visual encoding. This framework has profoundly influenced subsequent research and remains a cornerstone of color design theory. Building on Bertin's work, MacEachren (2004) further connected visual variables to cognitive and perceptual processes, emphasizing the alignment between color choices, data characteristics, and map-reading tasks.

Building on the above theoretical foundations, researchers have later sought to translate these color theories into practical guidelines. Brewer (1994) systematically investigated

color use in maps and proposed practical guidelines that relate color schemes to data measurement levels and mapping purposes. These principles were later operationalized through widely adopted tools and templates, such as ColorBrewer (Brewer et al. 2003; Harrower and Brewer 2003). Subsequent studies have examined perceptual aspects of color design, including brightness contrast, class discriminability, and visual consistency, demonstrating their influence on map readability and user interpretation (Brewer et al. 1997; Brychtová & Çöltekin, 2015; Ware 2019).

With the development of information technology, recent work has begun to formalize color knowledge through rule-based systems and interactive tools, integrating insights from perception, cartography, and design practice (Christophe 2011). Yang et al. (2025a); Wei et al. (2026a) even used this knowledge for a large language model to support interactive color design in practice. Together, these studies establish color design as a mature and theoretically grounded area of cartographic research, while also highlighting the diversity of design choices observed in practice.

3. Layout structure.

Layout design addresses the spatial organization of the map elements within a bounded page or screen (Jia et al. 2024; Yang et al. 2026). Classical cartographic literature has long emphasized layout principles derived from visual perception

and graphic design, including visual hierarchy, balance, alignment, and Gestalt grouping (Dent et al. 2008; Slocum et al. 2009; MacEachren 2004). These principles aim to ensure that the main map remains the dominant visual focus while supporting elements are arranged to facilitate efficient information access without competing for attention.

In addition to the above qualitative guidelines, researchers have increasingly sought to formalize and quantify layout quality. Early efforts modeled visual balance using mechanical analogies or geometric measures (Ma et al. 2013), while later studies proposed metrics for hierarchy, compactness, and alignment to evaluate layout effectiveness (O'Donovan et al., 2014). On the generative side, several approaches have attempted to translate expert layout knowledge into computational rules or optimization procedures, including case-based recommendation systems (Wei et al. 2017), explicit legend grouping rules (Qin and Li 2017), and integer programming-based layout optimization (Dayama et al. 2020).

More recently, data-driven approaches have gained prominence in layout research. Inspired by advances in document and interface design, learning-based models have been applied to capture layout patterns from large-scale datasets, such as LayoutGAN and LayoutVAE (Li et al. 2019; Jyothi et al. 2019). Within cartography, MapLayNet (Yang et al. 2025b) represents an early effort to learn map layout representations in a data-driven manner by modeling map elements and their spatial relationships as graphs and employing weakly supervised graph neural networks to capture global-local layout patterns from large collections of ubiquitous maps.

In summary, prior research has established comprehensive design principles for map elements, color usage and layout organization. However, most existing studies focus on prescribing design guidelines, developing tools, or analyzing limited collections of exemplary maps. Large-scale, longitudinal analyses that examine how these design principles are applied in academic publishing remain scarce.

Data-Driven Analysis of Maps

With the rapid expansion of digital mapping resources and the development of intelligent techniques, conducting large-scale data-driven studies of maps has become increasingly feasible in recent years (Robinson et al. 2017). GeoAI research has demonstrated that deep learning and related methods can effectively support large-scale geographic image analysis and mapping tasks by leveraging abundant digital map and spatial data sources (Li and Hsu 2022; He et al. 2025; Wei et al. 2026b).

Early data-driven studies on maps primarily treated map images as visual inputs for recognition-oriented tasks,

focusing on scanned cartographic products and thematic classification. Representative datasets compiled annotated map images into coarse thematic categories, such as political, physical, or topographic maps, to support research on text extraction, symbol recognition, and map-type classification (Mandal et al. 2014). With the adoption of deep convolutional neural networks, subsequent datasets expanded in scale and were used to train models that classify maps into different thematic or geographic types, such as terrain, urban, or land-use maps (Zhou et al. 2018).

As data-driven methods matured, research began to extend toward generative and segmentation-based tasks. Paired datasets linking cartographic renderings with aerial or satellite imagery enabled image-to-image translation between different map representations, exemplified by the Pix2Pix Maps dataset (Isola et al. 2017). In parallel, segmentation-oriented datasets derived from USGS topographic maps or historical cartographic archives were developed to support automated extraction of roads, hydrography, and other geographic features (Saeedimoghaddam and Stepinski 2020; Kramm et al. 2025). The EPFL works on historical cartography assembled ADHOC Records (771,561 catalog records aggregated from 38 institutions across 11 countries) and ADHOC Images (99,715 digitized map images), enabling computational investigations of cartographic evolution at an unprecedented scale (Petitpierre 2025). VisTaxa developed a hierarchical taxonomy (51 taxa) by manually coding 400 historical visualization images and then scaling labels to the full OldVisOnline collection (13,511 images) via similarity-based and zero-shot CLIP-assisted prediction (Zhang et al. 2025). While these efforts substantially advanced automated map digitization and feature extraction, they continued to conceptualize maps mainly as carriers of geographic information, with limited attention to design characteristics.

More recent studies have increasingly recognized maps as structured visual documents composed of multiple cartographic elements arranged according to design conventions. For example, CartoMark integrates map classification, text annotation, and style-related tasks within a unified benchmark (Zhou et al. 2024). The ubiMap and ubiMap-l datasets further provide element-level annotations across thousands of maps, enabling structural layout analysis and supporting learning-based models for layout retrieval and representation learning (Yang et al. 2025b). Complementary to element- and layout-centric approaches, recent fragment-based studies have also explored the visual language of maps by decomposing map images into elementary graphical units and quantifying long-term stylistic evolution across large historical corpora (Petitpierre et al. 2024). In addition to structural annotations, behavioral and perception-oriented datasets have also emerged. Eye-tracking datasets that record users' visual attention during map reading provide empirical

evidence of how different map designs and layouts influence perception and cognitive processing (He et al. 2023).

In summary, despite these advances, most existing map image datasets remain task-driven, with primary objectives centered on classification, segmentation, text detection, or retrieval. Cartographic design principles, particularly the color design and layout structure, and their large-scale empirical evaluation, are often addressed only implicitly or not at all. As a result, comprehensive, longitudinal analyses on examining how map design characteristics evolve across large academic map corpora are still required.

Dataset Construction

Journal Selection and Corpus Scope

To construct a representative corpus for analyzing long-term map design practices, we selected a total of sixteen authoritative journals closely related to cartography and geographic information science, including eight Chinese-language journals and eight English-language journals. The selection was based on four criteria: (1) strong disciplinary relevance to cartography, geographic information science, or closely related spatial sciences; (2) long-term publication records covering a substantial portion of the study period; (3) recognition within their respective academic communities; and (4) accessibility of complete article archives for large-scale data collection. The journal information is shown in Table 1.

For each selected journal, we collected all research articles published between 1990 and 2020. Although several journals were founded after 1990, they were still included in the corpus to ensure comprehensive coverage of the field. These include two Chinese journals: Journal of Geo-Information Science (founded in 1996) and Journal of Remote Sensing (founded in 1997); and three English journals: Transactions in GIS (founded in 1997), Geoinformatica (founded in 1997), and Journal of Maps (founded in 2005). While publication records for most journals are continuous within the selected period, Transactions in GIS represents a notable exception: although founded in 1997, no articles were published in 1998, resulting in a one-year gap. The founding year of each journal is provided in Table 1. These years are not used as matching criteria between Chinese- and English-language journals, as this study does not aim to compare journal histories but to examine map design characteristics within the same analysis period. Although the selected journals vary in founding years, both groups are analyzed within the identical publication window (1990–2020), ensuring that comparisons focus on design patterns and temporal evolution rather than journal age.

Table 1 Overview of selected journals and article counts in the corpus

Language	Journal name and its index in this study	Year founded	Article count
Chinese	Acta Geographica Sinica (地理学报) -A	1934	3704
	Science China Earth Sciences (中国科学: 地球科学)-B	1950	3633
	Bulletin of Surveying and Mapping(测绘通报)-C	1955	8798
	Acta Geodetica et Cartographica Sinica (测绘学报)-D	1957	3090
	Geomatics and Information Science of Wuhan University (武汉大学学报(信息科学版)) -E	1957	5101
	Science of Surveying and Mapping(测绘科学)-F	1976	4661
	Journal of Geo-Information Science(地球信息科学学报)-G	1996	1947
	Journal of Remote Sensing(遥感学报) -H	1997	2535
	Total in Chinese		33,469
English	Annals of the American Association of Geographers-I	1911	2516
	Professional Geographer - J	1949	2142
	The Cartographic Journal - K	1964	994
	Cartography and Geographic Information Science - L	1974	1063
	International Journal of Geographical Information Science - M	1987	2583
	Transactions in GIS - N	1997	1192
	Geoinformatica - O	1997	560
	Journal of Maps - P	2005	1213
	Total in English		12,263
	Total		45,732

The eight Chinese-language journals are fully open access and were downloaded directly from their official publishing platforms. English-language journals were accessed and downloaded through the Web of Science database. The number of articles collected from each journal is summarized in Table 1. In total, the corpus consists of 45,732 articles across the sixteen journals.

Automatic Map Extraction and Map Element Identification

Vision-Language-Based Document Understanding and Thematic Map Identification

Maps in academic journals are distributed across heterogeneous PDF formats. While some articles provide digitally generated figures with explicit structural metadata, others are scanned documents in which maps are embedded as rasterized images, especially in some early publications. To robustly extract map images from both cases, we adopt a hybrid parsing and vision-language-based identification strategy. The process of map extraction is shown in Fig. 2.

Map extraction via VLM and manual verification

Automatically Extracted Maps via VLM ($N=58112$)

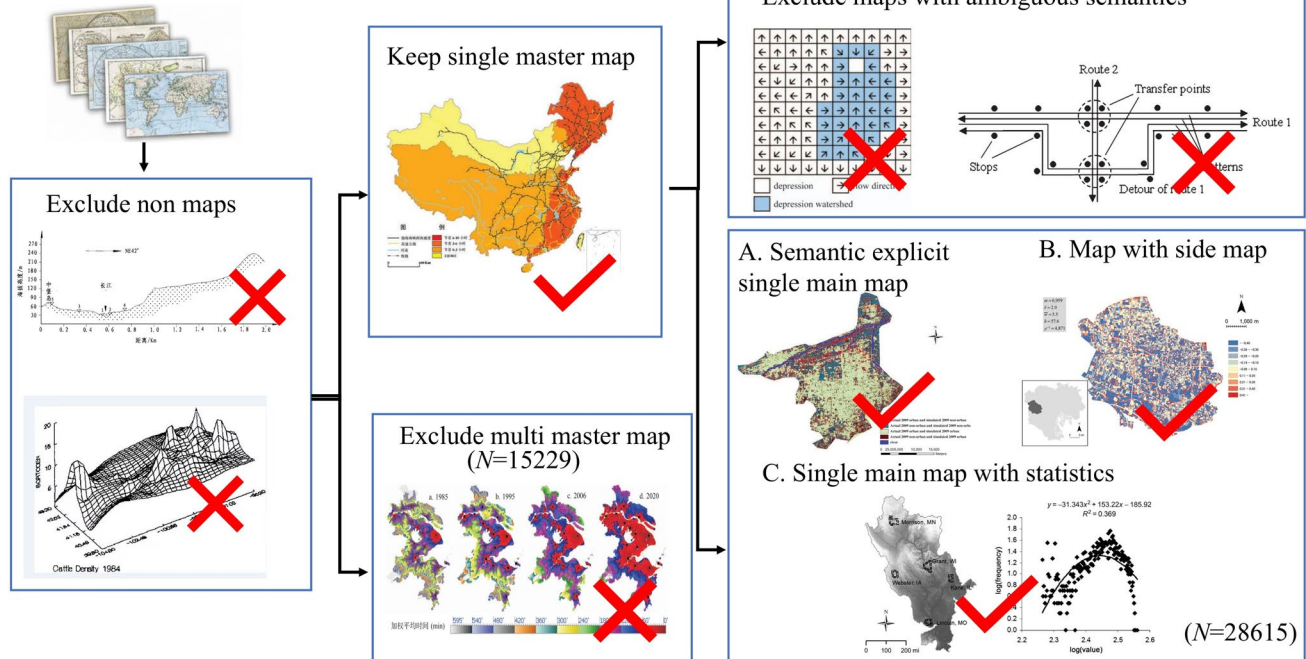


Fig. 2 Workflow for extracting maps from article PDF files. A total of 58,112 potential maps are initially identified using a vision-language model (VLM) from the PDF versions of articles. Manual verification is then conducted to filter and refine the results, yielding 28,615 maps

1. PDF parsing and vision-language-based map identification.

The maps from the article PDF files are identified with two stages: first, extracting images from PDF files, and second, identifying thematic maps from those images. Initially, we employ MinerU-10B, a vision-language-based document understanding model developed by OpenDataLab (Wang et al. 2024b), to parse PDF files and recover their internal visual and textual structures, including text blocks, tables, figures, and embedded images. This step ensures that we can recover candidate figure images regardless of whether the PDF is digitally authored or scanned. The associated caption text for each extracted image is also retained.

Once the images are extracted, we proceed to identify thematic maps using a locally deployed Qwen 7B vision-language model, one of the most powerful open-source vision models developed by Alibaba (Wu et al. 2025). Specifically, a two-stage prompting strategy is employed. The first stage involves the model distinguishing thematic maps from other non-thematic figures, such as charts, diagrams, photographs, and schematic illustrations. In the second stage, the model further filters out typical or ambiguous cases that may remain after the initial classification, ensuring a more refined selection of maps. Through this process, we obtain 58,112 potential map images, though some of these figures visually resemble

maps but are not suitable for design analysis, such as map-based statistical charts, heavily abstracted cartograms. Thus, a manual verification process is then performed.

2. Manual verification.

This manual verification process serves two primary purposes: (1) retaining only maps with a clearly defined single dominant main map, and (2) excluding maps with ambiguous or unclear semantic content. The requirement of a single dominant main map arises from the methodological assumptions of the subsequent element detection and layout analyses, which depend on a clearly identifiable primary map region for consistent measurement and comparison.

For the first task, we filter out images with multiple main maps. If an image contains multiple subfigures and each subfigure can be treated as an independent and complete map, it is extracted as a separate map. However, if the subfigures cannot function independently, the entire image is excluded from further analysis. In the second task, we exclude maps that primarily serve as conceptual or schematic illustrations, such as principle-oriented diagrams or abstract illustrative maps. Additionally, maps whose semantic meaning or geographic reference is unclear are also discarded. After this manual identification and filtering process, 28,615 maps are retained for further analysis.

Computer Vision-Based Cartographic Element Detection and Segmentation

To enable fine-grained analysis of map design, we then identify the cartographic elements in the maps collected in Sect. "Vision-language-based document understanding and thematic map identification" and explicitly separate the main map from surrounding elements. This distinction supports two subsequent analyses in this study: layout analysis based on element arrangement and color analysis conducted exclusively on the main map. Accordingly, this process consists of three steps: map element categorization, map element identification, and main map extraction.

1. Map element categorization.

Based on established cartographic design literature (Tait 2018; Jia et al. 2024), 10 common map elements that frequently appear in academic map layouts are adopted in this work, as defined in Table 2.

2. Map element identification.

To identify and localize map elements at scale, we adopt an annotation-and-detection pipeline similar to that used in the VisTaxa dataset (Zhang et al. 2025). A subset of representative maps (1,625 images) was manually annotated

according to the taxonomy defined in Table 2. Each map was independently labeled by two annotators, and disagreements were resolved through discussion to reach consensus. The resulting annotations were used to fine-tune a YOLOv8-OB object detection model, a state-of-the-art real-time detector supporting oriented bounding boxes, pre-trained on a large-scale dataset (Mitrev et al., 2025). The trained model was evaluated on a held-out validation set and the key evaluation metrics reached satisfactory levels: mAP@0.5 stabilized around 0.83, and mAP@0.5:0.95 stabilized around 0.68. The model was then applied to the full corpus to automatically extract the positions and extents of map elements in a consistent manner. During the element detection stage, 4,687 extracted map images were identified as lacking a recognizable main map and were therefore excluded from subsequent analysis, leaving 23,928 maps for further processing. Figure 3(b) illustrates representative examples of element identification across different thematic maps, where oriented bounding boxes are used to localize major components, including the main map, legend, scale bar, and other auxiliary elements.

3. Main map extraction.

Based on the oriented bounding boxes of the main map obtained during element identification, we further refine the main map regions using a segmentation-based approach. Specifically, the detected bounding box of the main map is used as a prompt for the Segment Anything Model (SAM), a foundation model for image segmentation with strong generalization capability across heterogeneous datasets (Kirillov et al. 2023), to generate a precise segmentation mask. Subsequent post-processing steps, including noise removal via Connected Component Analysis (Rosenfeld and Pfaltz 1966), are applied to retain only the primary geographic region while eliminating spurious fragments. The resulting main map masks serve as the basis for subsequent analyses of layout structure and color design, which focus exclusively on the map body. As shown in Fig. 3(c), the segmentation results accurately isolate the geographic regions from surrounding elements across maps with different layouts and visual styles.

Dataset Overview

As described above, 28, 615 academic thematic maps are extracted from 45,732 research articles following the automatic extraction and manual verification procedures outlined in Sect. "Journal selection and corpus scope". Subsequently, based on the map element detection and main map extraction process introduced in Sect. "Automatic map extraction and map element identification", 4,687 maps lacking

Table 2 Types of map elements considered in this study and their definitions

Element	Description
Title	Textual element that provides the main subject or theme of the map, usually placed in a prominent position to guide the reader's attention.
Main map	The primary geographic area depicted, serving as the central focus of the layout.
Legend	A reference box explaining the meaning of symbols, colors, or patterns used in the main map.
Scale bar	A graphical or numerical indicator that expresses the ratio between map distance and real-world distance.
Inset map	A supplementary map embedded within the layout, often used to show location context, enlargements, or global positioning.
Chart	The visualizations of statistical data, such as charts, graphs, or other forms of data representation designed to support the map's narrative.
Descriptive text	Explanatory or descriptive textual content (excluding the title), used to clarify background, context, or additional information.
North arrow	A directional indicator pointing to geographic north, assisting orientation.
Picture	Non-cartographic images (e.g., photos, diagrams) added to enrich the visual narrative of the map.
Table	Structured tabular information included within the layout, typically providing supporting statistical or categorical data.

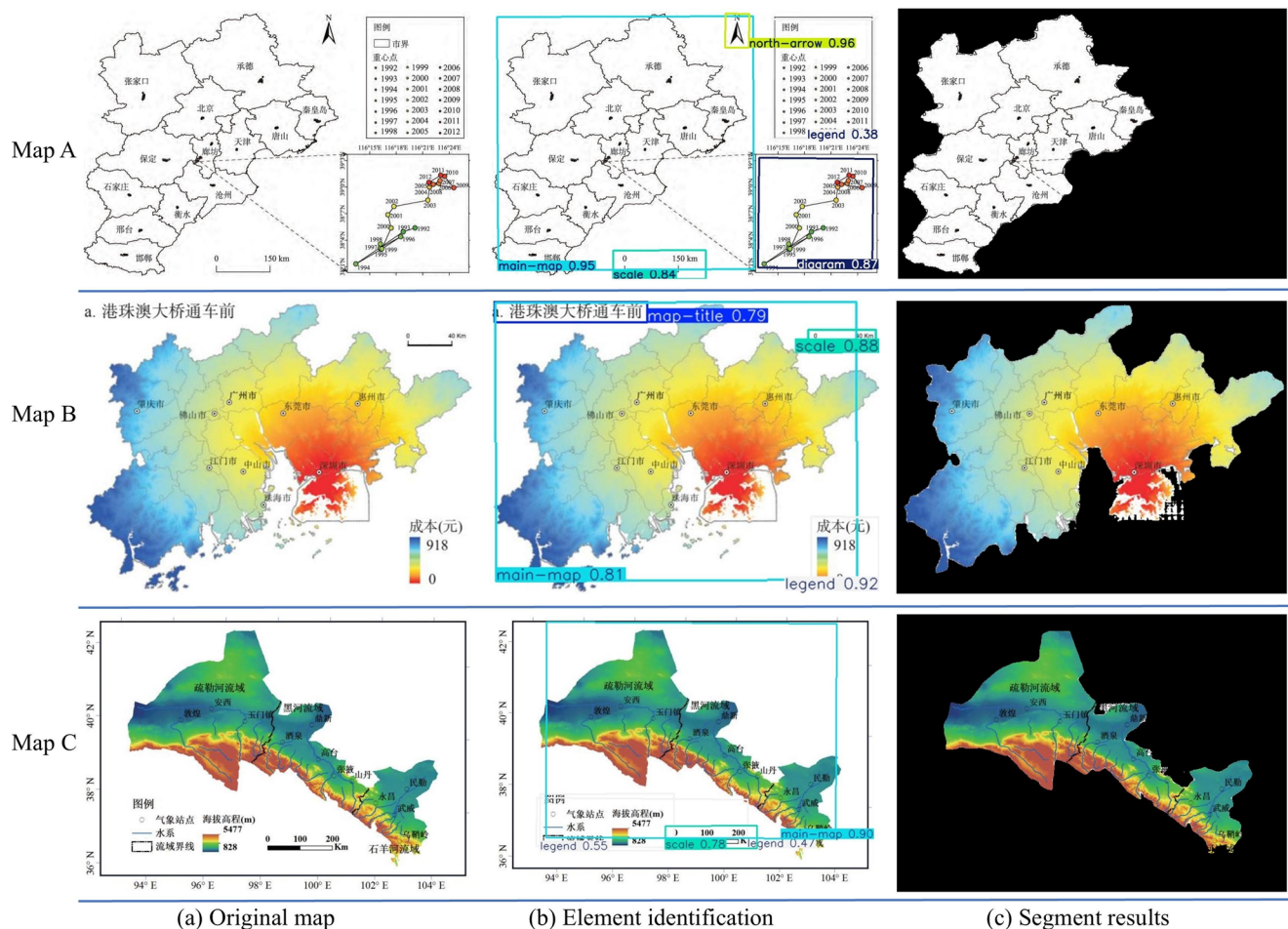


Fig. 3 Workflow of map element identification and main map extraction. (a) Original map; (b) Detected map elements; (c) Extracted main map region after segmentation

a recognizable dominant main map are removed, resulting in a final dataset of 23,928 maps for subsequent analysis. Table 3 summarizes the distribution of these 23,928 maps across journals and reports the corresponding map-to-article ratios (N/C).

As shown in Table 3, we can observe that Chinese-language journals contribute 15,971 maps, accounting for approximately two-thirds of the dataset, while English-language journals contribute 7,957 maps. This difference in absolute map counts largely reflects the underlying distribution of research articles, as Chinese-language journals collectively publish a substantially larger number of articles over the study period. When normalized by article count, we can observe that English-language journals exhibit a higher average map density ($N/C=0.65$) than Chinese-language journals ($N/C=0.48$), indicating a stronger tendency to include cartographic figures per article in English-language publications. For the variation across individual journals, we can observe that journals with an explicit cartographic focus, such as *The Cartographic Journal* ($N/C=1.39$) and

Journal of Maps ($N/C=0.95$), display the highest map densities, reflecting their emphasis on map contributions. In contrast, more method- or application-oriented journals, such as *Bulletin of Surveying and Mapping* ($N/C=0.16$) and *Professional Geographer* ($N/C=0.22$), show considerably lower ratios, where maps typically play a supporting rather than central role.

While Tables 1 and 3 summarize the aggregated distribution of articles and extracted maps across different journals, they do not reveal how academic publishing and map usage have evolved over time. Therefore, Fig. 4 further presents the temporal trajectories of article publication, map extraction, and map-to-article ratios from 1990 to 2020, providing a longitudinal perspective on changes in academic map practices. As shown in Fig. 4, Chinese- and English-language academic journals exhibit distinct evolutionary patterns in terms of publication volume and map usage.

In Chinese-language journals, both publication volume and map usage increased substantially, particularly after the 2000s. This indicates a rapid expansion of academic output

Table 3 Distribution of academic maps across journals (1990–2020)

Language	Journal index	Number of maps (<i>N</i>)	<i>N/C</i>
Chinese	A	3820	1.03
	B	4144	1.14
	C	1397	0.16
	D	718	0.23
	E	1202	0.24
	F	1651	0.35
	G	1832	0.94
	H	1207	0.48
	Total in Chinese	15,971	0.48
English	I	1331	0.53
	J	478	0.22
	K	1385	1.39
	L	420	0.40
	M	1687	0.65
	N	1225	1.03
	O	280	0.50
	P	1151	0.95
	Total in English	7957	0.65
	Total	23,928	0.52

Note: *N/C* represents the average number of thematic maps per article, calculated as the number of extracted maps (*N*) divided by the total number of articles (*C*) in each journal

and increasing integration of maps into scientific communication during this period. This significant increase may be attributed to China's economic and technological boom in the 21st century, which led to the widespread adoption of digital tools, advancements in GIS and cartography, and a broader shift towards map-based data visualization in academic research.

By contrast, English-language journals exhibit a more gradual evolutionary trajectory, reflecting relatively stable publication patterns and map usage practices over the study period. This more gradual rise could be a result of English-language journals already having well-established publishing infrastructures and cartographic standards earlier on,

limiting the scale of growth compared to Chinese journals, which experienced a period of catch-up after technological advancements became more accessible.

Regarding the map-to-article ratio in Fig. 4(b), English-language journals exhibit fluctuations in this ratio over time, but it remains relatively stable at a moderate level (about 0.6). This indicates that, while the number of maps increased, the ratio of maps to papers did not show dramatic changes, possibly due to the already established norms of academic publishing where maps are integral but not over-represented. On the other hand, Chinese-language journals show a gradual increase in the map-to-article ratio (from 0.3 to 0.6), reflecting a growing emphasis on maps in academic articles. This rise in the ratio could be due to the increasing importance of visual data representation in Chinese academic publishing, as well as the growing use of maps to convey complex spatial information in various fields of study.

Design Dimensions for Analysis

Based on the two research questions introduced in Sect. "Introduction", we define three analytical dimensions to systematically characterize academic map design: map elements, color design, and layout structure. Map elements are identified and described in Sect. "Automatic map extraction and map element identification". Thus, this section mainly describes the definitions and metrics for color design (Sect. "Color design") and layout structure (Sect. "Layout structure"), respectively.

Color Design

Color design plays a central role in map interpretation by shaping visual tone, perceptual differentiation, and overall

Temporal distribution of the academic articles and maps

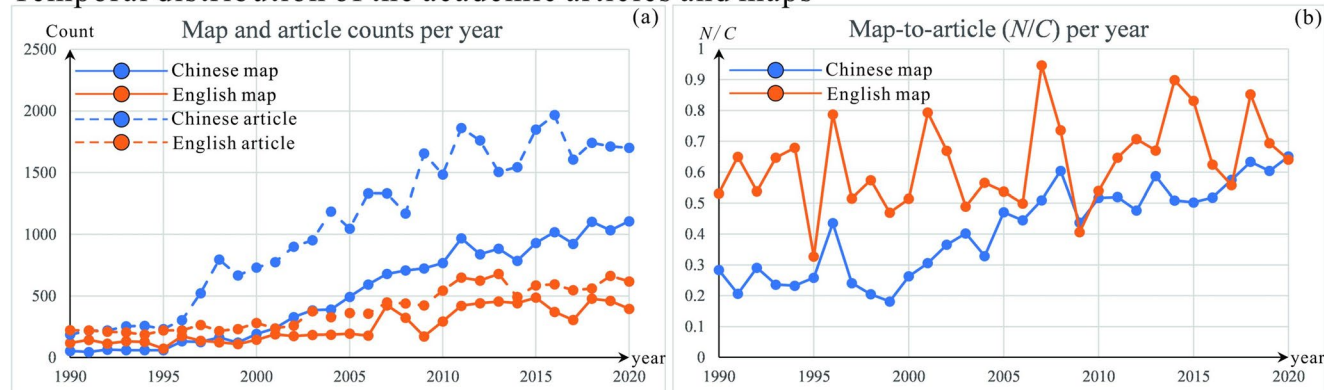


Fig. 4 Temporal evolution of academic publications, extracted maps, and map-to-article ratios in Chinese- and English-language journals (1990–2020)

readability (Brewer 2016). In this study, color analysis is conducted exclusively on the main map area, excluding legends and surrounding elements, to ensure that measured color properties reflect intrinsic map design rather than explanatory components. We characterize color design using two complementary principles: color tone and color complexity (Table 4). Color tone describes the intrinsic chromatic attributes of the map, including hue, saturation, and brightness, capturing the overall visual appearance of map colors. Color complexity characterizes the richness and organizational diversity of color usage, reflecting both the number of colors employed and their distribution across the map.

Layout Structure

Layout structure describes the spatial organization of the main map and auxiliary elements within a page (Tait 2018). In this study, we characterize layout structure using four complementary principles: hierarchy, visual balance, compactness, and alignment (Table 5). Hierarchy describes the relative visual prominence of the main map compared with surrounding elements. Visual balance captures the spatial equilibrium of elements across the page. Compactness reflects the efficiency of space usage in arranging map components, while alignment characterizes the consistency of element positioning along shared axes.

Statistical Analysis and Results

Cross-Language Differences in Academic Map Design

To answer RQ1, we examine whether Chinese- and English-language academic maps exhibit different design patterns across map elements, color design, and layout structure.

Cross-Language Differences in Map Elements

The cross-language differences in map elements are analyzed from three perspectives: (1) element quantity per map, (2) element type usage frequency, and (3) element co-occurrence patterns. To account for differences in total map counts between Chinese and English-language journals, all analyses are conducted on a proportional basis to ensure comparability. The analysis results are shown in Fig. 5; Table 6.

In terms of the number of elements per map, Fig. 5(a) shows that Chinese and English academic maps exhibit highly similar overall distributions. In both groups, maps with a small number of elements dominate. Specifically, maps containing one to three elements account for 79.3% of Chinese maps and 75.2% of English maps. The single-element and two-element cases alone represent 55.6% of Chinese maps and 54.4% of English maps, indicating that concise layouts are the prevailing design choice in both language contexts.

When summarized at the aggregate level, English-language maps contain a slightly higher average number of elements per map than Chinese-language maps (English: 2.59 elements per map; Chinese: 2.53 elements per map). While English maps show a slightly higher proportion of maps with four or more elements (approximately 24.8% compared to 20.7% in Chinese maps), this difference appears mainly in the long tail of the distribution and does not alter the overall similarity of element-count patterns between the two groups.

In terms of the element type usage, Fig. 5(b) shows that the relative frequencies of most cartographic elements are remarkably consistent across Chinese and English journals. For example, legends and scale bars are the most frequently used auxiliary elements in both groups. Legends appear in 45.48% of Chinese maps and 50.02% of English maps (much higher), while scale bars appear in 30.33% and 33.09%, respectively. North arrows show almost identical

Table 4 Evaluation metrics for map color design

Principle	Indicator	Description and definition	Source
Color tone	H_{main}	H_{main} is the dominant hue of the map.	Brewer (2016)
	S_{ave}	S_{ave} is the average saturation value.	
	B_{ave}	B_{ave} is the average brightness value.	
Color complexity	B_{con}	B_{con} is the statistical variation of brightness values across the map, defined	Cao et al. (2025)
		$asBcon = \sqrt{\frac{1}{N} \sum_{i=1}^N (B_i - B_{ave})^2}$ B_i represents the brightness value of the i -th pixel	
	N_{hue}	N_{hue} is the total number of distinct hues used in the main map, representing the basic richness of the color palette. To avoid counting marginal or noise colors, only hues accounting for at least 5% of the total pixel proportion in the main map are included in the calculation.	
	E_{hue}	E_{hue} is an entropy-based measure of hue distribution, and is defined as $E_{\text{hue}} = \sum_{i=1}^{N_{\text{hue}}} p(h_i) \log_2 p(h_i)$, where $p(h_i)$ represents the portion of pixels of hue h_i	

Table 5 Evaluation metrics for map layout quality

Principle	Indicator	Description and definition	Source
Hierarchy	D_{hier}	D_{hier} measures the positional deviation of the main map by calculating the Euclidean distance between its centroid and the visual center of the page. Horizontal and vertical offsets are first normalized by page width and height, respectively, so that both $D_{horizontal}$ and $D_{vertical}$ fall within the interval $[-1, 1]$. The overall deviation (D) is computed as: $D = \sqrt{D_{horizontal}^2 + D_{vertical}^2}$. After normalization, D is rescaled to the range $[0, 1]$, where 0 indicates perfect centering, and larger values represent increasing deviation from the visual center.	Dent et al. (2008)
Compactness	R_{map}	R_{map} measures the occupation of the main map relative to the total page and is defined as $R_{map} = A_{map}/A_{page}$, where A_{map} and A_{page} are the areas of the main map and the entire page, respectively. The ratio range is $[0, 1]$, with larger values indicating better compactness.	Wei et al. (2017)
Alignment	$R_{horizontal}$ and $R_{vertical}$	Alignment measures the structural consistency of layout elements and is defined as the proportion of misaligned sight lines relative to the total number of sight lines in the horizontal and vertical directions. $R_{horizontal} = \frac{C_{horizontal}}{L_{horizontal}}$, $R_{vertical} = \frac{C_{vertical}}{L_{vertical}}$ where $C_{horizontal}$ and $C_{vertical}$ are the number of misaligned horizontally and vertically sight lines, and $L_{horizontal}$ and $L_{vertical}$ are the total number of sight lines in the corresponding directions. The ratio range is $[0, 1]$, with smaller values indicating better alignment.	O'Donovan et al. (2014)
Visual balance	$B_{horizontal}$ and $B_{vertical}$	Visual balance is evaluated by using weighted contributions of map elements, considering both horizontal (top-bottom) and vertical (left-right) balance to ensure layout stability. The balance index is defined as the normalized difference between the visual weights on each side: $B = \frac{W_1 - W_2}{W_1 + W_2}$. Where W_1 and W_2 represent the total visual weights of elements on the two opposing sides (top-bottom and left-right). Thus, it has two visual balances in top-bottom ($B_{horizontal}$) and left-right ($B_{vertical}$), and falls within $[-1, 1]$. 0 indicates perfect equilibrium.	Wei et al. (2017)

usage rates (16.72% in Chinese maps and 16.55% in English maps). Other elements, such as titles, inset maps, pictures, tables, and charts, remain relatively infrequent in both datasets, with usage rates generally below 20%. Although English maps exhibit slightly higher frequencies for descriptive text (3.27% vs. 1.72%) and Table (0.60% vs. 0.43%), these elements account for only a small fraction and thus represent localized rather than structural differences.

In terms of element co-occurrence patterns, it is implemented via the Apriori algorithm (Huang et al. 2000), and the results are shown in Table 6. Table 6 further confirms the strong similarity in compositional conventions. In both Chinese-language maps and English-language maps, the most common element combination is the pairing of the main map with a legend, occurring in 62.29% of Chinese maps and 70.60% of English maps. The inclusion of a scale bar together with the main map is also common in both groups (41.54% in Chinese maps and 46.71% in English maps). Higher-order combinations, such as the joint presence of legend, scale bar, and north arrow, follow the same rank ordering across languages, although their absolute frequencies are moderately higher in English maps. These differences indicate that English maps tend to adopt slightly more fully specified combinations, but they do not alter the dominant co-occurrence shared by both groups.

Cross-Language Differences in Color Design

The cross-language differences in color design are analyzed based on the six indicators in Table 4. All color features are extracted using OpenCV in the HSB color space to ensure computational consistency. Hue values are classified according to the OpenCV hue scale and grouped into ten discrete categories: black, gray, white, red, orange, yellow, green, cyan, blue, and purple (Cao et al. 2025). Furthermore, we conduct Mann-Whitney U tests for the five quantitative indicators to assess cross-language differences, including S_{ave} , B_{ave} , B_{con} , N_{hue} , and E_{hue} . The statistical results, together with the distributional comparisons, are presented in Fig. 6.

1. Color tone.

Figure 6(a) shows that white is the main dominant hue in both Chinese-language maps and English-language maps, accounting for approximately 43% of Chinese maps and 40% of English maps. Gray tones account for roughly 25% in both datasets. Chromatic hues such as red, orange, yellow, green, and blue each individually account for less than 10%, indicating that academic maps in both contexts rely primarily on neutral or low-saturation hues.

For average saturation (S_{ave}) in Fig. 6(c), English-language maps exhibit significantly higher values than

Statistics of detected cartographic elements

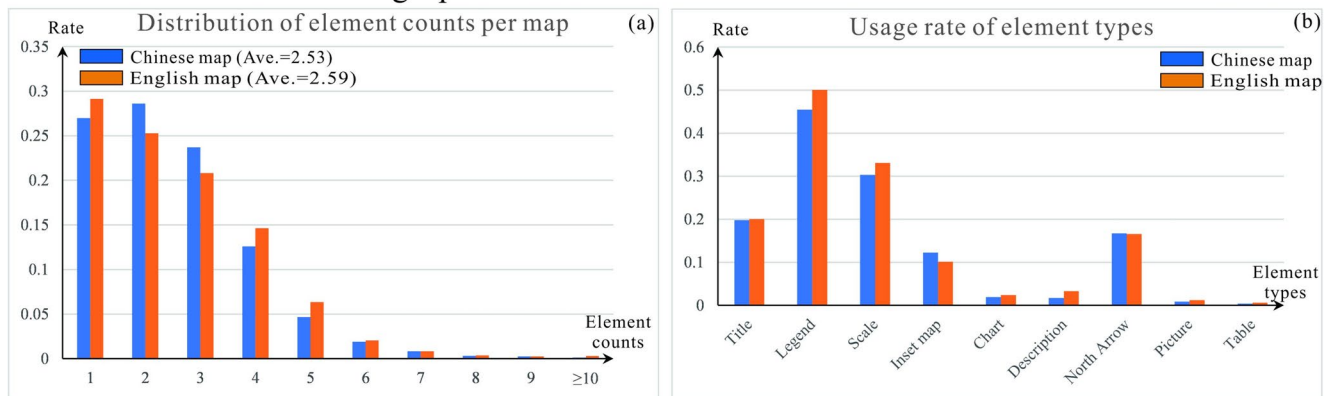


Fig. 5 Statistics of detected cartographic elements in the dataset

Table 6 Co-occurrence of map elements in Chinese- and English-language academic maps

Cooc- currence Type	Chinese		English	
	Element Combination	Rate (%)	Element Combination	Rate (%)
Top-2	Legend	62.29%	Legend	70.60%
	Main map		Main map	
	Scale bar	41.54%	Scale bar	46.71%
Top-3	Main map		Main map	
	Map title	27.14%	Legend	31.86%
	Main map		Scale bar	
Top-3	Scale bar	23.33%	Scale bar	31.86%
	Legend		Legend	
	Main map		Main map	
Top-3	Legend	13.51%	Legend	17.83%
	Main map		Main map	
	North arrow		North arrow	
Top-3	Map title	12.94%	Map title	17.14%
	Legend		Legend	
	Main map		Main map	

Chinese-language maps ($p=0.000***$). The median saturation for Chinese maps is 0.020, whereas the median for English maps is 0.075, indicating that English maps tend to employ relatively more saturated color schemes. However, despite this statistically significant difference, the overall saturation levels in both Chinese-language maps and English-language maps remain low. In both datasets, approximately 75% of maps have average saturation values below 0.25, suggesting that low-saturation or muted palettes dominate academic cartographic design. At the same time, the upper tails of the distributions indicate the presence of a small number of maps with markedly higher saturation values, reflecting occasional adoption of more vivid or bold color schemes. These results suggest that while English-language maps show a stronger tendency toward chromatic expressiveness, both Chinese-language maps and English-language maps largely adhere to restrained and academically conservative tonal conventions.

For average brightness (B_{ave}) in Fig. 6(c), Chinese-language maps display slightly higher values than English-language maps. The median brightness is 0.843 for Chinese maps and 0.836 for English maps. Although the statistical test indicates that the difference is significant ($p=0.005**$), the magnitude of the difference remains small, reflecting only a subtle tonal variation between the two groups. Importantly, both Chinese-language maps and English-language maps exhibit generally high brightness levels, with approximately 75% of maps concentrated in the upper brightness range, reflecting a dominant preference for light tonal compositions. This suggests that academic maps in both contexts commonly adopt bright backgrounds and high overall luminance.

2. Color complexity.

Figure 6(b) reveals a statistically significant difference in hue richness (N_{hue}) between the two language groups ($p=0.000***$). While maps using two hues dominate both groups (44.8% in Chinese and 38.1% in English), English-language maps show higher proportions in the three-hue (27.2% vs. 21.6%) and four-hue categories (13.2% vs. 11.2%). English maps also exhibit slightly longer tails in higher hue-count categories, indicating greater chromatic diversity. However, despite these differences, the overall distribution remains concentrated in low hue-count categories. In both Chinese-language maps and English-language maps, maps using no more than four hues account for over 85% of the total, demonstrating a shared tendency to avoid excessive chromatic complexity. This suggests that academic maps in both contexts favor restrained palettes, maintaining controlled hue variation while allowing limited flexibility in chromatic richness.

For brightness contrast (B_{con}) in Fig. 6(c), no statistically significant difference is observed between the two groups ($p=0.761$). The median values are nearly identical, at 0.169

Comparison of color design indicators

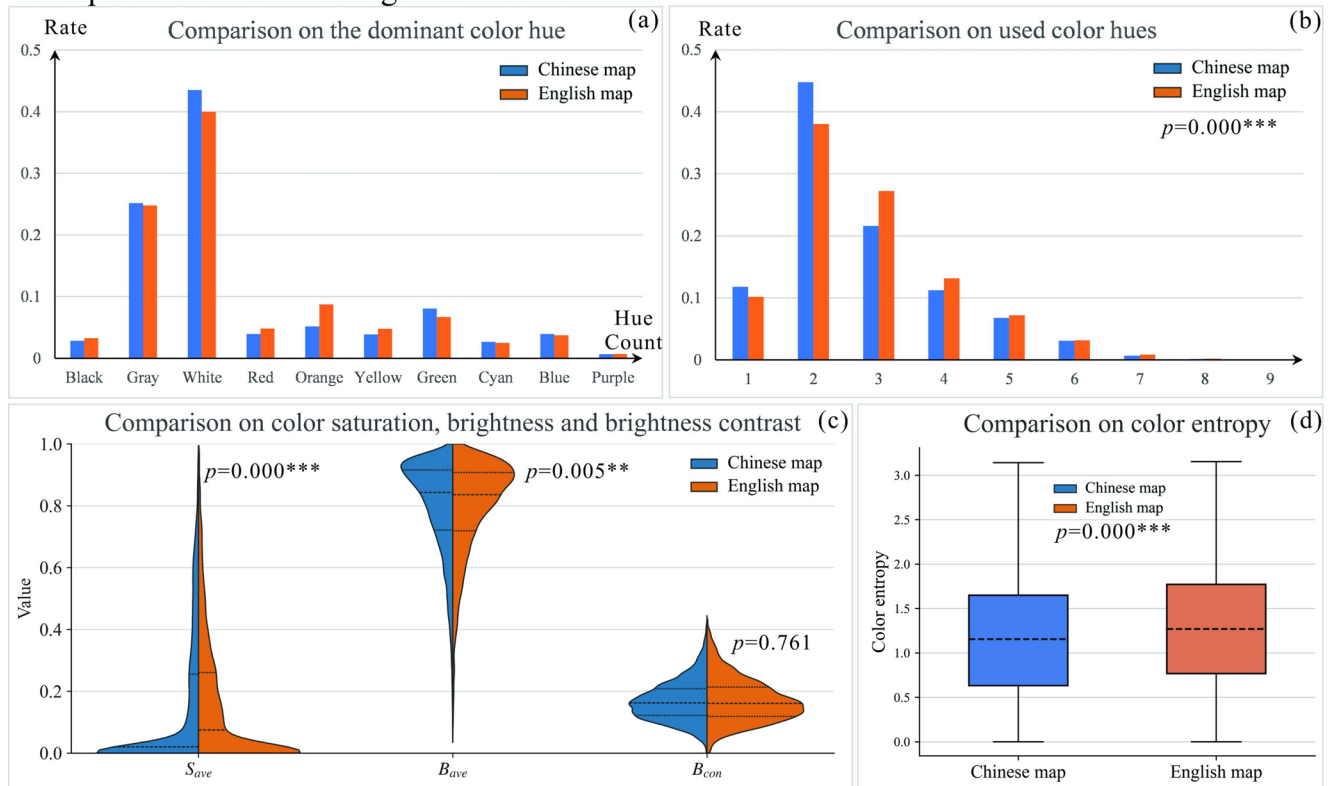


Fig. 6 Cross-language comparison of color design indicators between Chinese- and English-language academic maps

for Chinese maps and 0.170 for English maps, and the violin distributions largely overlap. Furthermore, the distributions show strong overlap, with almost all maps exhibiting brightness contrast values below 0.4, and approximately 75% of maps concentrated below 0.2 in both datasets. These results indicate that academic maps in both language contexts consistently employ low levels of tonal contrast, avoiding strong luminance differentiation across spatial units. This pattern suggests a shared design preference for moderate brightness variation, likely to maintain visual coherence and reduce perceptual distraction rather than to emphasize dramatic tonal segmentation.

For color entropy (E_{hue}) in Fig. 6(d), a statistically significant cross-language difference is observed ($p=0.000***$), with English-language maps exhibiting higher values overall. The mean entropy is 1.270 for English maps and 1.155 for Chinese maps, indicating relatively greater chromatic diversity in English-language publications. However, the overall distribution remains concentrated in lower entropy ranges in both groups. Approximately 75% of maps exhibit entropy values below 1.8, suggesting generally limited chromatic dispersion. This finding is consistent with the N_{hue} results, where the majority of maps employ no more than four distinct hues. In addition, a noticeable proportion of maps exhibit entropy values close to zero. This corresponds

to the presence of single-hue maps, which account for about 10% in both Chinese and English datasets. Taken together, while English-language maps display moderately higher color entropy, both Chinese-language maps and English-language maps mainly adopt restrained color distributions, reinforcing the controlled palette usage observed in the hue-count analysis.

Cross-Language Differences in Layout Structure

The cross-language differences in layout structure are analyzed based on the six indicators in Table 5. The Mann-Whitney U tests are conducted to examine cross-language differences for each quantitative indicator. The results are shown in Fig. 7. As shown in the figure, the overall layout structures of Chinese and English academic maps are highly comparable. Although several indicators reach statistical significance, the magnitudes of the differences are consistently small, and the distributions exhibit substantial overlap.

Specifically, for hierarchy (D_{hier}) in Fig. 7(a), its mean value is 0.101 for Chinese maps and 0.108 for English maps, indicating nearly identical centering practices. The statistical test confirms that the difference is not significant ($p=0.926$). In both Chinese-language maps and English-language

Comparison of layout structure indicators

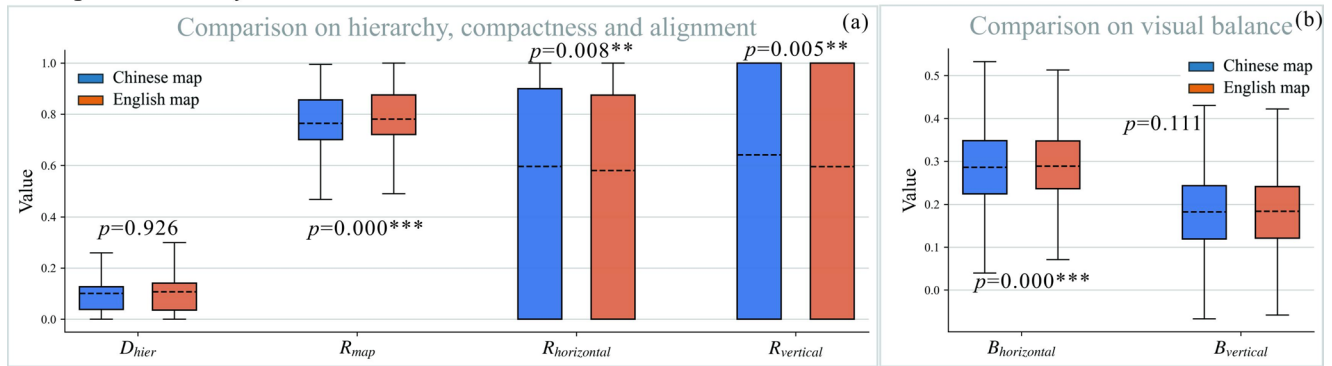


Fig. 7 Cross-language comparison of layout structure indicators between Chinese- and English-language academic maps

maps, the distributions are concentrated in the lower deviation interval (<0.2), demonstrating that academic maps consistently prioritize centered composition.

For compactness (R_{map}) in Fig. 7(a), a statistically significant difference is observed ($p=0.000***$), yet the magnitude of the difference remains limited. The mean value is 0.765 for Chinese maps and 0.781 for English maps, corresponding to a gap of only 0.016. Importantly, the distributions reveal strong concentration in higher compactness ranges. In both Chinese-language maps and English-language maps, nearly all maps exhibit R_{map} values above 0.5, and approximately 75% of maps exceed 0.7, indicating that the main map typically occupies a large proportion of the page. Although English-language maps show slightly higher compactness on average, the substantial distributional overlap suggests that both groups adopt a similarly main-map-dominant layout structure.

For alignment in Fig. 7(a), inconsistency remains comparable across languages. For horizontal alignment ($R_{horizontal}$), the mean is 0.596 (Chinese) versus 0.580 (English) ($p=0.008**$). For vertical alignment ($R_{vertical}$), the mean is 0.641 (Chinese) versus 0.595 (English) ($p=0.005**$). While these differences are statistically detectable, the absolute gaps (0.016 and 0.046, respectively) remain small relative to the full range [0,1], and the interquartile ranges substantially overlap, indicating only modest cross-language variation. Moreover, the distributions span a wide range, with a considerable number of maps exhibiting values close to 0 (indicating high alignment quality), alongside many maps with relatively high inconsistency values. This dispersion suggests that alignment is not uniformly enforced across academic maps. Instead, it appears to function as a flexible design dimension rather than a rigid rule.

For visual balance in Fig. 7(b), the differences are minimal. For horizontal visual balance ($B_{horizontal}$), the mean value is 0.286 (Chinese) and 0.289 (English) ($p<0.001$), indicating no statistically significant difference. For vertical balance ($B_{vertical}$), the mean value is 0.182 (Chinese) and

0.184 (English) ($p=0.111$). Although horizontal balance reaches statistical significance, the difference in mean values is less than 0.003, which is also practically negligible. Across both Chinese-language maps and English-language maps, the balance indices are predominantly positive, with mean values of approximately 0.28 for $B_{horizontal}$ and 0.18 for $B_{vertical}$. These values indicate a modest but consistent compositional tendency slightly upward in vertical placement and mildly asymmetric in horizontal distribution. Nevertheless, all values remain relatively close to zero, indicating that despite these subtle tendencies, academic maps in both language contexts continue to prioritize overall equilibrium and balanced composition.

Temporal Evolution of Academic Map Design

To answer RQ2, we investigate how academic map design has evolved from 1990 to 2020 and whether the two linguistic groups follow similar or divergent trajectories.

Temporal Evolution of Map Elements

To investigate the temporal evolution of map elements from 1990 to 2020, we analyze both the overall richness of map elements (i.e., the average number of elements per map) and the temporal changes in key auxiliary elements, including legends, scale bars, and north arrows, and further divide the study period into three stages: 1990–2000, 2001–2010, and 2011–2020. Spearman's rank correlation tests are conducted between publication year and each element-related indicator to quantify their long-term monotonic trends. As shown in Fig. 8, all indicators exhibit significant increasing trends in both Chinese- and English-language academic maps, although their evolutionary patterns vary across different developmental stages.

During the early stage (1990–2000), Chinese- and English-language maps showed clear differences in their level of cartographic completeness. English-language maps

generally contained more map elements, with an average element count of 2.29 compared with 2.03 for Chinese-language maps during this period (Fig. 8a). Similar differences were observed for auxiliary elements. The average legend usage rates were 0.37 and 0.23 for English- and Chinese-language maps, respectively (Fig. 8b), while the corresponding values for scale bar usage were 0.25 and 0.23 for English- and Chinese-language maps, respectively (Fig. 8c). North arrow usage remained relatively low in both groups, although English-language maps showed slightly higher usage (0.06 vs. 0.04; Fig. 8d). Overall, the early stage was characterized by relatively limited use of supporting cartographic elements, with English-language maps exhibiting slightly higher structural completeness.

During the transition stage (2001–2010), both language groups experienced continuous improvements in map element configuration. The average number of elements increased to 2.39 for English-language maps and 2.19 for Chinese-language maps (Fig. 8a). The usage of legends also continued to rise, reaching average values of 0.48 and 0.38 for English- and Chinese-language maps, respectively (Fig. 8b). Scale bar usage showed a moderate increase, with average rates of 0.29 and 0.28 for English- and Chinese-language maps (Fig. 8c). Meanwhile, north arrow usage increased more substantially in English-language maps than

in Chinese-language maps, with average rates of 0.15 and 0.06, respectively (Fig. 8d). These results indicate that the transition period was characterized by a general improvement in map element completeness, although some differences remained between the two groups.

During the recent stage (2011–2020), Chinese- and English-language maps gradually converged toward similar configurations. The average element count reached 2.77 for Chinese-language maps and 2.80 for English-language maps (Fig. 8a), indicating comparable levels of overall element richness. Similarly, legend usage increased to 0.51 and 0.55 for Chinese- and English-language maps, respectively (Fig. 8b), while scale bar usage reached 0.32 and 0.38 (Fig. 8c). North arrow usage also increased substantially in both groups, with average rates of 0.23 for Chinese-language maps and 0.21 for English-language maps (Fig. 8d). These results suggest that by the end of the study period, both groups had adopted relatively similar patterns of map element configuration, characterized by approximately three elements per map and more frequent inclusion of essential auxiliary components.

Overall, the temporal evolution of map elements demonstrates a gradual transition from initially different levels of cartographic completeness toward increasing convergence

Temporal evolution of map element configuration

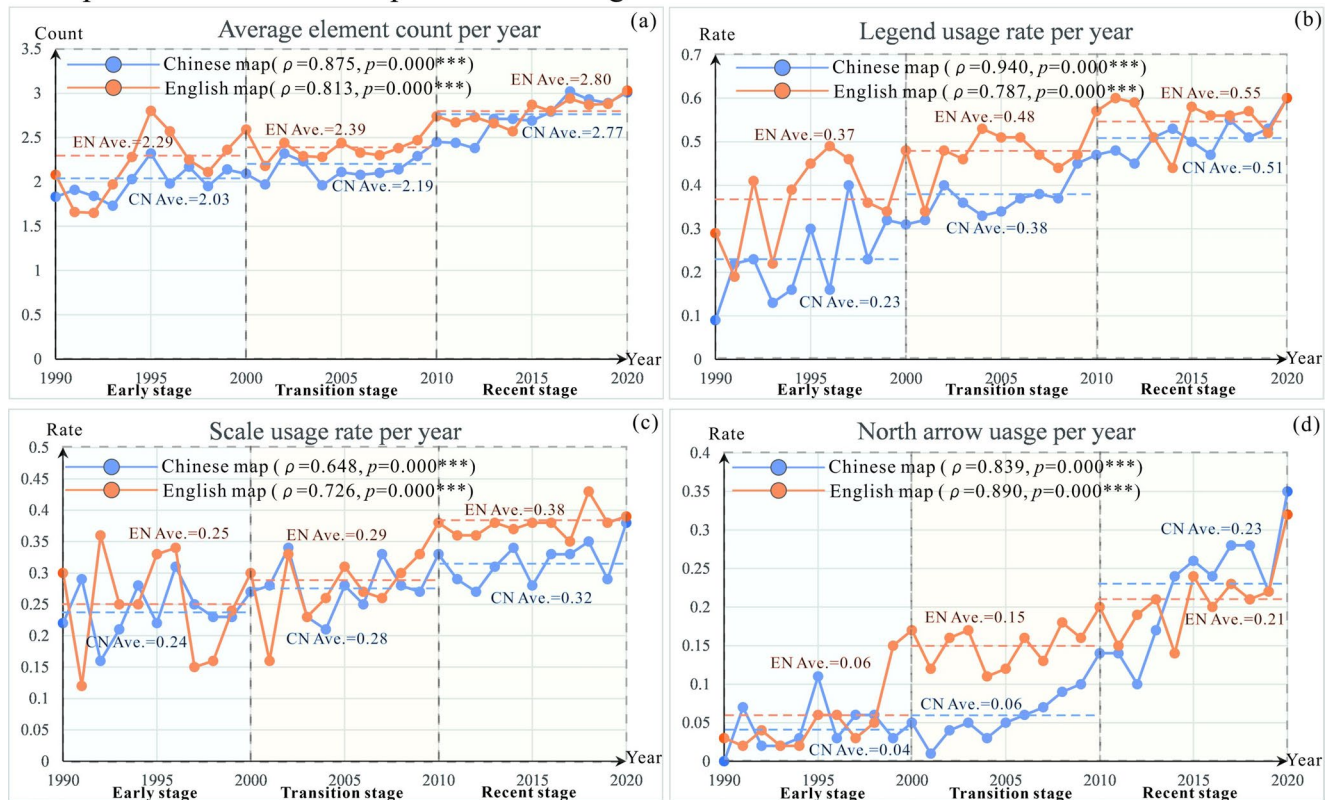


Fig. 8 Temporal evolution of map element configuration in Chinese- and English-language academic maps (1990–2020). The shaded areas indicate three developmental stages: Early stage (1990–2000), Transition stage (2001–2010), and Recent stage (2011–2020)

between Chinese- and English-language academic maps. Spearman correlation results further confirm significant positive temporal trends for all indicators, including average element count ($\rho = 0.875$ for Chinese maps; $\rho = 0.813$ for English maps), legend usage ($\rho = 0.940$; $\rho = 0.787$), scale bar usage ($\rho = 0.648$; $\rho = 0.726$), and north arrow usage ($\rho = 0.839$; $\rho = 0.890$). These findings indicate that academic map design has progressively moved toward more complete and standardized configurations over the past three decades.

Temporal Evolution of Color Design

To investigate how color design practices evolved from 1990 to 2020, we analyze the annual trajectories of six color-related indicators (Table 4) and further divide the study period into three stages: 1990–2000, 2001–2010, and 2011–2020. As discussed in Sect. "Cross-language differences in color design", black, white, and gray dominate the overall hue composition of academic maps in both language groups. Therefore, for dominant hue analysis, we focus on the proportion of chromatic dominant hues, excluding achromatic colors, to characterize changes in the adoption of color tones over time. Spearman's rank correlation tests are conducted to quantify monotonic relationships between publication year and each color indicator. Overall, both Chinese-language and English-language maps show a consistent transition from relatively restrained color schemes toward richer and more diverse color expressions, although English-language maps generally exhibit higher levels of chromatic diversity throughout the study period.

During the early stage (1990–2000), academic maps in both language groups were characterized by limited chromatic usage and relatively simple color structures. The proportion of chromatic dominant hues remained low, with average values of only 0.06 for Chinese-language maps and 0.07 for English-language maps (Fig. 9a). Similarly, S_{ave} was low, with both groups showing an average value of approximately 0.03 (Fig. 9b), indicating a strong preference for achromatic and low-saturation color schemes. N_{hue} was also relatively limited, averaging 2.26 and 2.38 for Chinese-language and English-language maps, respectively (Fig. 9e), while E_{hue} remained below 1.0 (0.78 and 0.94, Fig. 9f). These results suggest that color design during this period was relatively conservative, likely influenced by traditional cartographic production methods and printing constraints.

During the transition stage (2001–2010), both groups experienced substantial increases in color richness and diversity. The usage rate of chromatic dominant hues increased markedly, reaching average values of 0.20 for

Chinese-language maps and 0.28 for English-language maps (Fig. 9a), representing more than a threefold increase compared with the early stage. S_{ave} also increased to 0.10 and 0.14, respectively (Fig. 9b), indicating a gradual shift toward more vivid color applications. Meanwhile, N_{hue} increased to 2.50 for Chinese-language maps and 2.74 for English-language maps (Fig. 9e), and color entropy increased to 1.02 and 1.23 (Fig. 9f). These changes indicate that academic maps began to adopt more complex color combinations during this period, corresponding with the widespread adoption of digital cartographic production and improved color reproduction technologies.

During the recent stage (2011–2020), the evolution toward richer color expression continued, but with more moderate changes in some indicators. The proportion of chromatic dominant hues further increased to 0.36 for Chinese-language maps and 0.43 for English-language maps (Fig. 9a), while average saturation reached 0.19 and 0.21, respectively (Fig. 9b). Color complexity also continued to increase, with N_{hue} reaching 2.86 for Chinese-language maps and 3.03 for English-language maps (Fig. 9e), and E_{hue} increasing to 1.28 and 1.41 (Fig. 9f). Compared with the early stage, these values demonstrate a clear increase in chromatic diversity and complexity over the three decades. However, the growth rate became relatively slower after 2010, suggesting that color design practices gradually approached a more stable stage.

In contrast to the increasing trends in chromatic diversity, brightness-related indicators show different evolutionary patterns. B_{ave} remained relatively stable throughout the entire period (Fig. 9c). Chinese-language maps maintained consistently high B_{ave} , decreasing slightly from 0.86 in the early stage to 0.78 in the recent stage, whereas English-language maps remained almost unchanged (0.81 to 0.79). This indicates that both groups consistently preferred bright background and tonal compositions despite increasing color diversity. B_{con} , however, exhibited a decreasing trend in both groups (Fig. 9d). B_{con} decreased from 0.22 to 0.16 for Chinese-language maps and from 0.22 to 0.15 for English-language maps, suggesting that recent academic maps tend to achieve visual differentiation through richer hue variations rather than stronger tonal contrasts.

Although the overall evolutionary trajectories are highly consistent, differences between Chinese-language and English-language maps remain evident. English-language maps generally show higher chromatic usage, saturation, hue diversity, and color entropy across all three periods. For example, in the recent stage, English-language maps achieved higher values in chromatic dominant hue usage (0.43 vs. 0.36), S_{ave} (0.21 vs. 0.19), N_{hue} (3.03 vs. 2.86), and E_{hue} (1.41 vs. 1.28). Statistical comparisons between the two language groups further confirmed these differences,

Temporal evolution of color design indicators

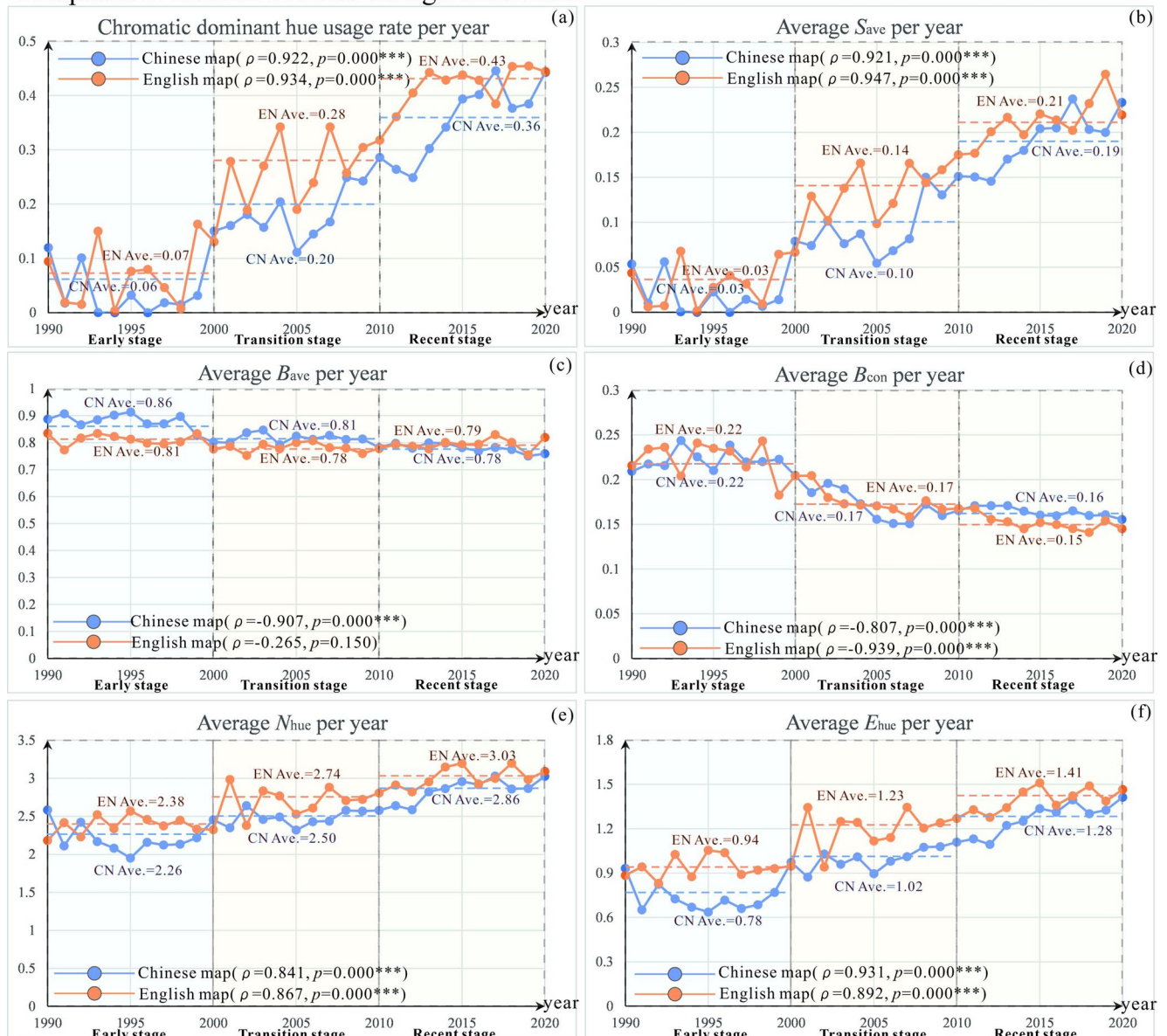


Fig. 9 Temporal evolution of color design indicators in Chinese- and English-language academic maps (1990–2020). The shaded areas indicate three developmental stages: Early stage (1990–2000), Transition stage (2001–2010), and Recent stage (2011–2020)

with English-language maps exhibiting significantly higher values for chromatic dominant hue usage, S_{ave} , N_{hue} , and E_{hue} ($p < 0.001$). In contrast, brightness-related indicators showed relatively smaller differences between the two groups, with B_{ave} remaining consistently high and B_{con} following similar decreasing patterns. These results suggest that although both cartographic communities have undergone a similar transition toward richer and more diversified color design, English-language maps tend to adopt more complex and diverse color strategies, whereas Chinese-language maps maintain relatively more conservative color expressions.

Temporal Evolution of Layout Structure

To investigate the temporal evolution of layout structure from 1990 to 2020, we analyze six annual layout indicators in Table 5 and further divide the study period into three stages: 1990–2000, 2001–2010, and 2011–2020. Spearman's rank correlation tests are conducted between publication year and each indicator to identify potential monotonic temporal trends. Overall, layout structure exhibits relatively stable temporal patterns compared with map elements and color design, although several indicators show gradual changes over the thirty-year period (Fig. 10).

Temporal evolution of layout structure indicators

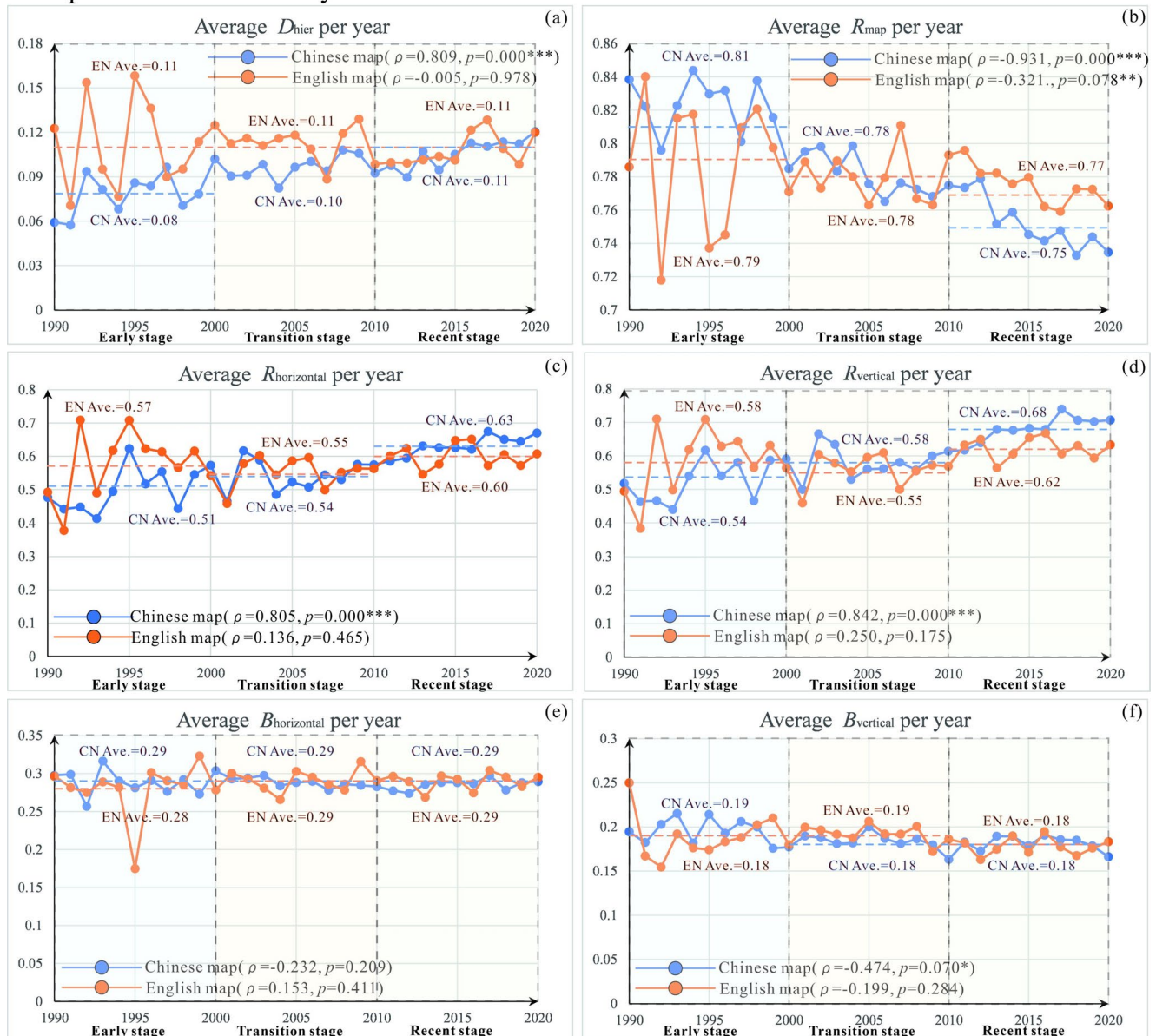


Fig. 10 Temporal evolution of layout structure indicators in Chinese- and English-language maps (1990–2020). The shaded areas indicate three developmental stages: Early stage (1990–2000), Transition stage (2001–2010), and Recent stage (2011–2020)

During the early stage (1990–2000), Chinese- and English-language maps already exhibited similar overall layout characteristics, with relatively centered and balanced compositions. The average D_{hier} values were 0.08 for Chinese-language maps and 0.11 for English-language maps, indicating limited deviation from centered layouts (Fig. 10a). Similarly, R_{map} remained relatively high in both groups, with average values of 0.81 and 0.79 for Chinese- and English-language maps, respectively (Fig. 10b). For alignment, the average $R_{horizontal}$ values were 0.51 and 0.57, while the corresponding $R_{vertical}$ values were 0.54 and 0.58 for Chinese- and English-language maps, respectively

(Fig. 10c and d). In contrast, the visual balance indicators remained much closer to equilibrium, with $B_{horizontal}$ averaging approximately 0.29 and 0.28 and $B_{vertical}$ averaging 0.19 and 0.18, respectively (Fig. 10e and f). These results suggest that fundamental layout principles, such as visual balance and centered composition, were already relatively stable at the beginning of the study period.

During the transition stage (2001–2010), most layout indicators remained relatively stable, while the main-map occupation ratio showed a gradual decline. Specifically, R_{map} decreased to average values of 0.78 for Chinese-language maps and 0.78 for English-language maps (Fig. 10b),

indicating that the relative proportion of the main map area gradually decreased as additional information components occupied more layout space. Meanwhile, D_{hier} remained around 0.10–0.11, and the visual balance indicators changed little, with $B_{\text{horizontal}}$ remaining around 0.29 and B_{vertical} around 0.18–0.19. Alignment indicators showed somewhat greater variation: $R_{\text{horizontal}}$ averaged 0.54 and 0.55, while R_{vertical} averaged 0.58 and 0.55 for Chinese- and English-language maps, respectively. These patterns indicate that although map layouts became slightly more information-dense, the overall compositional principles remained largely unchanged.

During the recent stage (2011–2020), layout structures continued to exhibit strong stability in hierarchy and visual balance, while more noticeable changes occurred in space occupation and alignment. The average R_{map} values decreased further to 0.75 for Chinese-language maps and 0.77 for English-language maps (Fig. 10b), representing the only consistent decreasing trend observed across both language groups. Visual balance also remained highly stable, with $B_{\text{horizontal}}$ averaging approximately 0.29 in both groups and B_{vertical} approximately 0.18. In contrast, the alignment indicators increased more noticeably in Chinese-language maps: $R_{\text{horizontal}}$ reached 0.63 compared with 0.60 in English-language maps, while R_{vertical} reached 0.68 compared with 0.62 (Fig. 10c and d). These results suggest that contemporary academic maps continue to maintain centered and balanced compositions, while undergoing incremental adjustments in space allocation and element alignment.

Although some indicators show statistically significant temporal trends, their practical effects on layout organization are limited. For example, D_{hier} in Chinese-language maps exhibits a significant positive correlation with publication year ($\rho = 0.809$, $p < 0.001$), whereas English-language maps show no significant trend ($\rho = -0.005$, $p = 0.978$). Similarly, both $R_{\text{horizontal}}$ ($\rho = 0.805$, $p < 0.001$) and R_{vertical} ($\rho = 0.842$, $p < 0.001$) increase significantly in Chinese-language maps, whereas the corresponding trends in English-language maps are not significant. However, these changes occur within a relatively narrow range, with D_{hier} values remaining approximately between 0.08 and 0.12 and R_{vertical} varying within the mid-range of the scale. Therefore, these statistically detectable changes indicate gradual adjustments in layout organization rather than fundamental shifts in design principles.

Overall, the temporal evolution of layout structure demonstrates a high degree of continuity from 1990 to 2020. Unlike map elements and color design, which experienced substantial changes over time, layout structure remained relatively stable, characterized by persistent visual balance, centered composition, and gradual reduction in main-map occupation ratio. The most evident temporal adjustments

occurred in space allocation and alignment, particularly in Chinese-language maps, rather than in the fundamental organization of the layout. These findings indicate that academic map layouts have evolved mainly through incremental adjustments rather than major structural transformations.

Discussion

Cross-Language Differences in Academic Map Design (RQ1)

Regarding RQ1, our results demonstrate that Chinese- and English-language academic maps exhibit differences in several design dimensions, including map element configuration, color usage, and layout organization. However, these differences should not be interpreted as fundamentally different cartographic design systems. Instead, they represent variations in how shared cartographic principles are implemented under different cultural, academic, and publishing contexts. Specifically, differences in color expression and visual organization may be associated with distinct aesthetic traditions and communication preferences, whereas differences in map elements may reflect variations in academic practices and publication environments.

One important difference is observed in color design. Chinese-language maps demonstrate a stronger preference for muted and neutral color schemes. As shown in Figs. 6 and 68.8% of Chinese maps are dominated by gray or white tones, and only 10.6% employ more than four hues. In contrast, English-language maps exhibit relatively greater chromatic diversity, with 64.8% dominated by gray or white tones and 11.4% using more than four hues. These differences may reflect the interaction between cultural aesthetics and cartographic traditions. Traditional Chinese visual culture has long emphasized harmony, balance, and restrained expression, which may partly explain the preference for relatively subtle color compositions in Chinese-language maps (Zong 1981). In contrast, Western visual traditions and modern cartographic practices have generally placed greater emphasis on perceptual differentiation, contrast, and categorical distinction, which are also central principles in thematic map color design (Bertin 1974; Brewer 1994; MacEachren 2004). Therefore, cross-language differences in color usage may arise from the combined influence of contextual aesthetic preferences and established cartographic practices.

Differences in visual organization are also reflected in layout structure. As shown in Fig. 7, Chinese- and English-language maps exhibit significant differences in compactness (R_{map}). Chinese academic maps tend to integrate multiple supporting components, such as legends, scale

bars, and descriptive texts, within a relatively limited map frame, resulting in more information-dense compositions. In contrast, English-language maps generally allocate relatively more space to the main map area and arrange auxiliary elements more independently. These differences may reflect different expectations regarding information presentation and visual communication. Although general cartographic principles emphasize visual hierarchy, balance, and efficient organization of map components (Dent et al. 2008; Slocum et al. 2009), their practical implementation may vary across different academic and cultural contexts.

Beyond cultural and visual traditions, academic publishing environments also influence how cartographic principles are translated into practice. Academic maps are produced within specific scholarly communication systems, where journal standards, editorial expectations, and disciplinary traditions shape design decisions (Robinson et al. 2017). For example, differences in map element configuration can be observed between Chinese- and English-language publications. In the early period, English-language maps contained more elements on average than Chinese-language maps (2.08 vs. 1.83 elements per map), and key auxiliary elements were also more frequently included, including legends (0.29 vs. 0.09) and scale bars (0.30 vs. 0.22). These differences suggest that variations in academic publishing practices and disciplinary conventions may have influenced the emphasis placed on cartographic completeness and supporting map components.

Overall, Chinese- and English-language academic maps exhibit both differences and shared characteristics. While contextual factors influence specific design choices, particularly in color expression, information organization, and element configuration, both groups follow common cartographic principles. Thus, the observed cross-language differences are better understood as contextual adaptations of shared cartographic conventions rather than separate design paradigms.

Temporal Evolution and Convergence of Academic Map Design (RQ2)

Regarding RQ2, our results reveal that academic map design underwent substantial but dimension-dependent evolution from 1990 to 2020. Across the three developmental stages, Chinese- and English-language maps generally followed similar directions of change, but the degree of convergence varied across design dimensions. The most evident convergence occurred in map element configuration, where the initial gap in structural completeness largely disappeared in the recent stage. Color design evolved in parallel toward richer chromatic expression, although English-language maps consistently retained greater color diversity. In contrast,

layout structure remained comparatively stable throughout the three decades, with changes occurring mainly in space allocation and alignment rather than in fundamental compositional principles. Thus, the overall temporal pattern is better characterized as increasing convergence within shared evolutionary directions, accompanied by persistent differences in some design characteristics.

The evolution of map elements provides the clearest evidence of convergence. During the early stage (1990–2000), English-language maps contained more elements on average than Chinese-language maps (2.29 vs. 2.03), and legends were also more frequently included (0.37 vs. 0.23). During the transition stage (2001–2010), element richness increased in both groups, reaching 2.39 elements per map for English-language maps and 2.19 for Chinese-language maps, while legend usage increased to 0.48 and 0.38, respectively. By the recent stage (2011–2020), however, the average element counts had become nearly identical (2.80 vs. 2.77), and the difference in legend usage had narrowed substantially (0.55 vs. 0.51). North-arrow usage showed an even more notable change: English-language maps had higher average usage during the transition stage (0.15 vs. 0.06), whereas the two groups became comparable in the recent stage (0.21 vs. 0.23). These patterns indicate that the earlier advantage of English-language maps in cartographic completeness was gradually reduced, suggesting increasing adoption of shared conventions concerning essential map components.

Color design followed a different pattern: the two groups evolved in the same direction, but cross-language differences persisted. During the early stage, both groups relied heavily on restrained color schemes, with chromatic dominant hues accounting for only 0.06 of Chinese-language maps and 0.07 of English-language maps. By the transition stage, these values had increased to 0.20 and 0.28, and by the recent stage to 0.36 and 0.43, respectively. N_{hue} similarly increased from 2.26 to 2.86 in Chinese-language maps and from 2.38 to 3.03 in English-language maps, while E_{hue} rose from 0.78 to 1.28 and from 0.94 to 1.41, respectively. At the same time, B_{con} declined from approximately 0.22 in both groups during the early stage to 0.16 and 0.15 in the recent stage. These synchronized changes suggest a shared transition toward richer hue differentiation combined with more moderate tonal contrast. However, unlike map element configuration, color design did not fully converge: English-language maps continued to exhibit higher chromatic usage, hue richness, and color entropy in the recent stage. Therefore, the temporal evolution of color design reflects parallel development with persistent stylistic differences, rather than complete convergence.

Layout structure exhibits an even stronger degree of continuity. Across all three stages, hierarchy deviation remained low, with average D_{hier} values changing only from 0.08

to 0.11 in Chinese-language maps and remaining around 0.11 in English-language maps. Visual balance was also remarkably stable: $B_{\text{horizontal}}$ remained around 0.28–0.29 and B_{vertical} around 0.18–0.19 across the three stages in both groups. The clearest temporal change concerned the occupation of the main map. R_{map} declined from 0.81 to 0.75 in Chinese-language maps and from 0.79 to 0.77 in English-language maps between the early and recent stages, indicating a gradual reduction in the proportion of page space occupied by the main map. Meanwhile, alignment-related indicators increased more noticeably in Chinese-language maps, particularly $R_{\text{horizontal}}$ (0.51 to 0.63) and R_{vertical} (0.54 to 0.68), whereas the corresponding changes were smaller in English-language maps. Taken together, these findings suggest that the fundamental spatial organization of academic maps—particularly centering and visual balance—has remained highly persistent, while incremental adjustments have occurred in space allocation and alignment as map compositions became more information-rich.

The stage-wise patterns also help explain why convergence is more pronounced in some dimensions than in others. Academic map production has been increasingly supported by shared digital infrastructures, including GIS software, digital cartographic tools, and standardized scientific publishing workflows (Goodchild 2007). These developments may have had a particularly strong influence on relatively explicit and operational design practices, such as the inclusion of legends, scale bars, and other cartographic elements, thereby contributing to the strong convergence observed in map element configuration. The substantial increase in Chinese-language map completeness from the early to the recent stage is consistent with this interpretation. Increasing internationalization of academic publishing may have further promoted common expectations concerning figure preparation and informational completeness.

At the same time, the persistence of cross-language differences in color design indicates that technological standardization does not necessarily eliminate all contextual variation. Although improved digital production and color reproduction capabilities enabled both groups to employ increasingly diverse palettes, choices concerning saturation, hue richness, and chromatic emphasis remain more flexible and may continue to reflect different visual traditions and publishing preferences. Similarly, the relative stability of layout indicators suggests that some basic compositional principles, such as centering and visual balance, were already well established at the beginning of the study period and therefore experienced less substantial transformation than map elements or color usage.

Overall, the thirty-year analysis indicates that Chinese- and English-language academic maps have followed broadly shared evolutionary directions, but convergence is

neither uniform nor complete. Map element configuration shows strong convergence from initially different levels of structural completeness; color design exhibits parallel enrichment while retaining moderate cross-language differences; and layout structure remains largely stable, with only incremental changes in compactness and alignment. These findings suggest that the long-term evolution of academic cartography reflects an interaction between increasingly shared technological and professional standards and persistent contextual preferences, producing greater standardization without eliminating all cross-language variation.

Implications for Academic Map Design

Although statistically detectable cross-language differences are observed in several indicators, the overall design structures of Chinese- and English-language academic maps are highly similar. Across map elements, color usage, and layout structure, both groups exhibit restrained chromatic strategies, moderate alignment flexibility, strong centering tendencies, and increasing structural completeness over time. These characteristics are consistent with established cartographic principles emphasizing effective visual encoding, perceptual clarity, and balanced composition (Bertin 1974; MacEachren 2004; Dent et al. 2008). More importantly, the longitudinal trajectories are broadly parallel but dimension-dependent: map-element configuration shows strong convergence, color design evolves toward greater chromatic richness while retaining cross-language differences, and layout structure remains comparatively stable with incremental changes in space allocation and alignment.

1. Color design.

Although English-language maps exhibit slightly higher saturation, hue richness (N_{hue}), and color entropy (E_{hue}), both Chinese-language maps and English-language maps overwhelmingly rely on neutral dominant hues, low overall saturation, high brightness (B_{ave}), and restrained tonal contrast (B_{con}). For example, more than 85% of maps in both datasets use no more than four hues, and approximately 75% of maps remain concentrated in low-saturation and high-brightness ranges. The temporal trajectories of most color indicators are broadly consistent: N_{hue} and E_{hue} increase in both language groups, while B_{con} declines. Brightness remains relatively high overall, although Chinese-language maps show a more evident decrease in B_{ave} over time than English-language maps.

These parallel trends suggest that academic color design is evolving along a shared path toward greater chromatic richness combined with moderated tonal differentiation, consistent with established recommendations for thematic

map color design emphasizing perceptual discrimination and controlled visual contrast (Bertin 1974; Brewer 1994; MacEachren 2004). For contemporary map design practice, these findings imply that color strategies in academic cartography are largely standardized across linguistic contexts. Designers may adopt restrained palette structures with moderate hue diversity, high-luminance backgrounds, and controlled brightness contrast, following established principles for effective thematic map communication (Brewer 1994; Ware 2019), without requiring language-specific adaptation.

2. Layout structure.

Layout indicators exhibit even stronger cross-language consistency. As shown in Sect. "Statistical analysis and results", hierarchy deviation (D_{hier}), compactness (R_{map}), alignment ($R_{\text{horizontal}}$ and R_{vertical}), and visual balance ($B_{\text{horizontal}}$ and B_{vertical}) display substantial distributional overlap between Chinese- and English-language maps. Although certain differences reach statistical significance (e.g., slightly higher R_{map} in English maps), the magnitude of these gaps remains small relative to their full scales. Temporal analysis further reinforces this structural similarity. From the early to the recent stage, R_{map} decreases from 0.81 to 0.75 in Chinese-language maps and from 0.79 to 0.77 in English-language maps, indicating a gradual reduction in the relative space occupied by the main map. Meanwhile, D_{hier} remains low at approximately 0.08–0.11, and both visual balance indicators remain nearly unchanged across the three stages. Alignment indicators show somewhat greater temporal adjustment, particularly in Chinese-language maps, but these changes do not constitute a fundamental reorganization of map layouts.

These patterns suggest that academic layout design is governed by persistent compositional principles—centered main maps, high main-map occupation ratios, near-equilibrium balance, and moderate alignment flexibility—regardless of language. These observations align with classical cartographic layout theories emphasizing visual hierarchy, balance, alignment, and effective organization of map components (Dent et al. 2008; Slocum et al. 2009; Tait 2018). For practical map design, the results therefore support the use of these shared structural principles as general guidance, while allowing flexible adjustments in space allocation and alignment according to specific information and publishing requirements.

Limitations and Future Work

While this study provides a longitudinal and multilingual examination of thematic map design evolution, several limitations should be acknowledged.

First, the dataset is restricted to academic journal publications between 1990 and 2020, and the majority of the

analyzed cases follow a single main-map layout structure. Although the selected journals represent authoritative sources in Chinese and English cartography, they do not encompass all forms of thematic maps. In particular, multi-main-map layouts, composite figures, and small-multiple designs have become increasingly common in recent years, especially in interdisciplinary and data-rich publications (Yang et al. 2026). The current study, therefore, does not explicitly model interactions among multiple coordinated map panels. As such, the findings mainly reflect conventions within single-map academic compositions rather than the full structural diversity of contemporary cartographic practice. Future research may extend the analytical framework to systematically examine multi-map configurations and explore whether similar convergence patterns persist in more complex layout systems.

Second, the quantitative indicators employed in this study focus on structural and chromatic dimensions, including element richness, hue diversity, hierarchical deviation, compactness, alignment, and balance. While these metrics capture important aspects of visual composition, other design dimensions (such as typography, symbol semantics, label density, annotation style, and thematic classification strategies) are not systematically measured. Future work could incorporate additional feature extraction methods to construct a more comprehensive representation of map design.

Third, although cross-language convergence is demonstrated across multiple indicators, the present analysis remains observational rather than causal. The study identifies synchronized trends but does not directly disentangle the relative influence of technological advances, editorial guidelines, software standardization, or broader globalization processes. Future research may combine policy analysis, editorial guideline comparison, or interviews with practitioners to further clarify the mechanisms underlying standardization.

Conclusion

This study investigates the longitudinal evolution of thematic map design in academic cartography and examines whether cross-language differences constitute enduring divergence or transitional variation. By constructing a multilingual dataset spanning three decades and operationalizing map design characteristics into quantitative indicators, the study provides large-scale empirical evidence on how academic map design develops over time. The results demonstrate that Chinese- and English-language academic maps share broadly similar evolutionary directions, although the degree of convergence varies across different design dimensions. Specifically, map element configuration exhibits the strongest convergence: initial differences in structural completeness, including the use of legends, scale bars, and other auxiliary components,

have substantially diminished over time. In contrast, color design shows parallel evolution rather than complete convergence. Both groups have gradually adopted richer chromatic expressions and more diverse palettes, while English-language maps consistently maintain higher levels of chromatic diversity and hue complexity. Layout structure remains the most stable dimension, with fundamental compositional characteristics such as centered hierarchy and visual balance remaining largely unchanged, accompanied mainly by incremental adjustments in main-map occupation and alignment. These findings suggest that academic map design has evolved within an increasingly shared cartographic framework shaped by common publishing practices, digital production technologies, and professional standards, while contextual differences remain evident in specific design choices, particularly color expression. Overall, the long-term evolution of academic cartography reflects a process of partial standardization rather than complete homogenization: shared conventions have become increasingly prominent, but variations associated with linguistic, cultural, and publishing contexts continue to influence specific aspects of map design.

Future work could focus on leveraging the constructed dataset to support downstream cartographic applications, such as data-driven map design recommendations, automated cartographic layout generation, and adaptive visualization systems. In particular, the large-scale and longitudinal nature of the dataset provides opportunities to bridge empirical analysis with practical map-making workflows, enabling the development of AI-assisted cartographic design tools.

Acknowledgements The authors would like to thank the editors and anonymous reviewers for their useful comments on the manuscript.

Author contributions Zhiwei Wei: Conceptualization, Methodology, Investigation, Writing - Original draft preparation, Supervision, Project administration, Funding Acquisition. Chenxi Song: Methodology, Data curation, Software, Validation, Visualization, Writing - Original draft preparation. Tazhu Wang: Methodology, Data curation, Software, Visualization, Writing - review editing. Wenjia Xu: Writing - Review & Editing, Supervision. Fan Wu: Writing - Review & Editing, Supervision. Hua Liao: Writing - Review & Editing, Supervision, Project administration, Funding Acquisition. Su Ding: Writing - Review & Editing, Supervision. Nai Yang: Writing - Review & Editing, Supervision, Project administration, Funding Acquisition.

Funding This work was supported by grants from the National Natural Science Foundation of China (No. 42501551, 42371455), Tobii China Innovation Initiative Project (TPI250407CN).

Data Availability The data that support the findings of this study are all openly available on Huggingface. The website: <https://huggingface.co/datasets/song12321/papermap>.

Declarations

Competing interest The authors declare no competing interest.

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